

Principle Component Analysis (PCA)

PCA: background

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)}$$

PCA: background

An example. We'll make up the covariance matrix for an imaginary 3 dimensional data set, using the usual dimensions x , y and z . Then, the covariance matrix has 3 rows and 3 columns, and the values are this:

$$C = \begin{pmatrix} cov(x, x) & cov(x, y) & cov(x, z) \\ cov(y, x) & cov(y, y) & cov(y, z) \\ cov(z, x) & cov(z, y) & cov(z, z) \end{pmatrix}$$

Matrix Algebra

- This serves to provide a background for the matrix algebra required in PCA.

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 11 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 8 \end{pmatrix} = 4 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Figure 2.2: Example of one non-eigenvector and one eigenvector

$$2 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 24 \\ 16 \end{pmatrix} = 4 \times \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

Figure 2.3: Example of how a scaled eigenvector is still an eigenvector

- Eigenvectors can only be found for *square* matrices. And, not every square matrix has eigenvectors. And, given an $n \times n$ matrix that does have eigenvectors, there are n of them. Given a 3×3 matrix, there are 3 eigenvectors.

Exercises

For the following square matrix:

$$\begin{pmatrix} 3 & 0 & 1 \\ -4 & 1 & 2 \\ -6 & 0 & -2 \end{pmatrix}$$

Decide which, if any, of the following vectors are eigenvectors of that matrix and give the corresponding eigenvalue.

$$\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Principle Component Analysis(PCA)

Steps Involved in the PCA

Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

Step 3: Calculate the eigenvalues and eigenvectors for the covariance matrix. & sort eigenvalues and their corresponding eigenvectors.

Step 4: Pick k eigenvalues and form a matrix of eigenvectors.

Principle Component Analysis(PCA): Example

- Data Set

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

Standardize the dataset: Subtract the mean

For PCA to work properly, you have to subtract the mean from each of the data dimensions. The mean subtracted is the average across each dimension. So, all the x values have $\bar{x}$ (the mean of the x values of all the data points) subtracted, and all the y values have $\bar{y}$ subtracted from them. This produces a data set whose mean is zero.

PCA

1. Standardize the Dataset

$$x_{new} = \frac{x - \mu}{\sigma}$$

$$\text{variance } \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

where:

x_i = the i^{th} data point

$\bar{x}$ = the mean of all data points

n = the number of data points

	f1	f2	f3	f4
μ =	4	3	3	3.4
σ =	3	1.58114	1.73205	2.30217

f1	f2	f3	f4
-1	-0.63246	0	0.26062
0.33333	1.26491	1.73205	1.56374
-1	0.63246	-0.57735	-0.17375
0.33333	0	-0.57735	-1.04249
1.33333	-1.26491	-0.57735	-0.60812

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

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2. Calculate the covariance matrix for the whole dataset

For Population

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

	f1	f2	f3	f4
f1	var(f1)	cov(f1,f2)	cov(f1,f3)	cov(f1,f4)
f2	cov(f2,f1)	var(f2)	cov(f2,f3)	cov(f2,f4)
f3	cov(f3,f1)	cov(f3,f2)	var(f3)	cov(f3,f4)
f4	cov(f4,f1)	cov(f4,f2)	cov(f4,f3)	var(f4)

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- Covariance matrix

	f1	f2	f3	f4
f1	0.8	-0.25298	0.03849	-0.14479
f2	-0.25298	0.8	0.51121	0.4945
f3	0.03849	0.51121	0.8	0.75236
f4	-0.14479	0.4945	0.75236	0.8



4.2 Eigenvalues and Eigenvectors

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Theorem 4.12. *The eigenvectors $x_1, \dots, x_n$ of a matrix $A \in \mathbb{R}^{n \times n}$ with n distinct eigenvalues $\lambda_1, \dots, \lambda_n$ are linearly independent.*

This theorem states that eigenvectors of a matrix with n distinct eigenvalues form a basis of $\mathbb{R}^n$.

Definition 4.13. A square matrix $A \in \mathbb{R}^{n \times n}$ is defective if it possesses fewer than n linearly independent eigenvectors.

A non-defective matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ does not necessarily require n distinct eigenvalues, but it does require that the eigenvectors form a basis of $\mathbb{R}^n$. Looking at the eigenspaces of a defective matrix, it follows that the sum of the dimensions of the eigenspaces is less than n . Specifically, a defective matrix has at least one eigenvalue λ_i with an algebraic multiplicity $m > 1$ and a geometric multiplicity of less than m .

Remark. A defective matrix cannot have n distinct eigenvalues, as distinct eigenvalues have linearly independent eigenvectors (Theorem 4.12). $\triangle$

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3. Calculate eigenvalues and eigen vectors.

- An **eigenvector** is a nonzero vector that changes at most by a scalar factor when that linear transformation is applied to it. The corresponding **eigenvalue** is the factor by which the eigenvector is scaled.
- Let **A** be a square matrix (in our case the **covariance matrix**), **v** a vector and λ a scalar that satisfies $Av = \lambda v$, then λ is called eigenvalue associated with eigenvector **v** of **A**.
Rearranging the above equation,

$$Av - \lambda v = 0 ; (A - \lambda I)v = 0$$

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- $Av - \lambda v = 0$; $(A - \lambda I)v = 0$
- Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if

$$\det(A - \lambda I) = 0$$

	f1	f2	f3	f4
f1	$0.8 - \lambda$	-0.25298	0.03849	-0.14479
f2	-0.25298	$0.8 - \lambda$	0.51121	0.4945
f3	0.03849	0.51121	$0.8 - \lambda$	0.75236
f4	-0.14479	0.4945	0.75236	$0.8 - \lambda$

Solving the above equation = 0

$$\lambda = 2.51579324 , 1.0652885 , 0.39388704 , 0.02503121$$

In general, once eigenvectors are found from the covariance matrix, the next step is to order them by eigenvalue, highest to lowest. This gives you the components in order of significance. Now, if you like, you can decide to *ignore* the components of lesser significance. You do lose some information, but if the eigenvalues are small, you don't lose much. If you leave out some components, the final data set will have less dimensions than the original.

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- **Eigenvectors:**

Solving the $(A-\lambda I)v = 0$ equation for v vector with different λ values:

$$\begin{pmatrix} 0.800000 - \lambda & -(0.252982) & 0.038490 & -(0.144791) \\ -(0.252982) & 0.800000 - \lambda & 0.511208 & 0.494498 \\ 0.038490 & 0.511208 & 0.800000 - \lambda & 0.752355 \\ -(0.144791) & 0.494498 & 0.752355 & 0.800000 - \lambda \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = 0$$

For $\lambda = 2.51579324$, solving the above equation using Cramer's rule, the values for v vector are

$$v_1 = 0.16195986$$

$$v_2 = -0.52404813$$

$$v_3 = -0.58589647$$

$$v_4 = -0.59654663$$

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- Going by the same approach, we can calculate the eigen vectors for the other eigen values. We can form a matrix using the eigen vectors.

e1	e2	e3	e4
0.161960	-0.917059	-0.307071	0.196162
-0.524048	0.206922	-0.817319	0.120610
-0.585896	-0.320539	0.188250	-0.720099
-0.596547	-0.115935	0.449733	0.654547

$$\lambda = 2.51579324, 1.0652885, 0.39388704, 0.02503121$$

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4. Pick k eigenvalues and form a matrix of eigenvectors
- If we choose the top 2 eigenvectors, the matrix will look like this:

e1	e2
0.161960	-0.917059
-0.524048	0.206922
-0.585896	-0.320539
-0.596547	-0.115935

This the final step in PCA, and is also the easiest. Once we have chosen the components (eigenvectors) that we wish to keep in our data and formed a feature vector, we simply take the transpose of the vector and multiply it on the left of the original data set, transposed.

$$FinalData = RowFeatureVector \times RowDataAdjust,$$

where *RowFeatureVector* is the matrix with the eigenvectors in the columns *transposed* so that the eigenvectors are now in the rows, with the most significant eigenvector at the top, and *RowDataAdjust* is the mean-adjusted data *transposed*, ie. the data items are in each column, with each row holding a separate dimension.